

Muhammad Wissam

Agentic AI SWE | Islamabad

As a Computer Science graduate with experience in Agentic AI and RAG systems, I believe strongly in the Agentic AI revolution in computing. Working on different AI systems has taught me that the actual bottleneck in any AI system today is the right execution, and I am willing to explore any niche possibility to achieve the best possible results. Spending time on things that actually matter is how I try to achieve my goals, and my personal motto is meaningful iteration towards a meaningful system.

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EXPERIENCE

Junior Software Engineer in AI

Synergy-IT Pakistan

07/2025 – Present

Achievements / Tasks

- Used the **Microsoft Agentic Framework** to develop an intelligent insurance claim filing system, reducing claim submission from **25+ screens** on average to **5 screens on average** — improving the customer experience by 5x and boosting customer retention.
- Developed and deployed an **AI CV Processing pipeline** to streamline the data flow of relevant applications to relevant databases, and implemented **RAG Search** on applicants' CV details to surface the best possible candidates — using **Azure AI Foundry** and **Azure Function Apps**.
- Developed and deployed a **Microsoft Teams Agentic Bot** that answers employee questions using the company's data documents with the help of the Agentic Toolkit.

SWE Intern

LMKR

07/2024 – 10/2024

- Deployed a **NetFlow monitoring system** using **ElastiFlow**, ingesting vectorized flow data through its **Elasticsearch** plugin.
- Packaged tools with **Docker** and **Docker Compose** to port them across different servers.

PROJECTS

TreeVerse (Final Year Project) | www.TreeVerse.qzz.io

- Built a vegetation monitoring web app using **Django REST Framework**, **React**, **PostgreSQL** and a **U-Net Model**. It enables access to interactive maps with **satellite imagery** and geospatial data. Users can **monitor deforestation** and **analyze vegetation** across 4 categories: crops, bushes, trees and barren land.

MLOps – End-to-End Pipeline

- A Machine Learning project with full end-to-end implementation of an **ML Orchestration pipeline**, with version control for data and models using **Jenkins**, **MLflow**, **Airflow**, **DVC** and **GitHub Actions**.

PixelFocus – Image Super Resolution

- An **image super-resolution app** using a hybrid **ResNet-U-Net-ESPCN** model to upscale images to high resolution. Designed with **TensorFlow** and **PyTorch** to achieve **2x image upscaling**.

FLEX – University Management System

- A university management web app developed using **ASP.NET MVC** and **MS SQL**, aimed at streamlining administration tasks like attendance and grading.

Video Generator – Prediction-Based Video Generation

- Built a locally deployed deep learning model using **Flask** to **generate a video** by **predicting the next 5 frames** from the previous 10 frames, using a **Vision Transformer** and **other models**.

SKILLS

- Programming Languages:** Python, C# (ASP.NET)
- AI & Agentic:** Microsoft Agentic Framework, Azure AI Foundry, MS Teams Agentic Toolkit, TensorFlow
- Databases & Search:** ChromaDB, Elasticsearch, MongoDB
- Cloud & Containers:** Docker, AWS Cloud
- MLOps & CI/CD:** Jenkins, MLflow, Airflow, GitHub Actions, DVC

ACHIEVEMENTS

Awarded a 100% Merit Scholarship for HSSC at Punjab College of Science, Blue Area, Islamabad.

CERTIFICATES

AWS Cloud Foundations

Gained a deep understanding of cloud computing with hands-on experience across **Amazon EC2, S3, RDS and IAM**.

AWS Cloud Data Pipeline Builder

Learned to build a scalable, **automated data pipeline** using AWS services for transformation and storage with **Amazon Redshift** and **AWS S3**.

AWS Cloud Web Application Builder

Learned to deploy a **serverless cloud-native web application** with services like **AWS Lambda** and **DynamoDB** for a secure, available and optimized app.

ORGANIZATIONS

FAST Literary Society

2022 – 2025

Served as **PR Manager** initially, followed by an **Information Secretary** role in FLS.

LANGUAGES

English

Full Professional Proficiency

Urdu

Native / Bilingual Proficiency

Turkish

Elementary Proficiency

Shina

Native / Bilingual Proficiency

INTERESTS

- Language Learning
- Geopolitics
- Hiking

EDUCATION

Bachelor's in Computer Science

FAST National University of Computer and Emerging Sciences, Islamabad

07/2021 – 06/2025

Courses

- Software Engineering
- MLOps
- Generative AI
- Operating Systems
- Data Structures
- Computer Networks